

PSY405 - Lesson 34

The Stimulus–Response (S–R) Theory of Dollard and Miller

Dollard and Miller tried to answer one main question:

| How does personality develop through learning?

Their answer was simple but powerful:

| Personality is a collection of learned habits.

And a **habit** is just:

| A learned connection between a stimulus (something you notice) and a response (what you do).

This is why their theory is called **Stimulus–Response (S–R) Theory**.



CORE IDEA: HABIT

Habit:

A **stable, learned association** between a **stimulus (cue)** and a **response**.



In simple English

A habit is when:

- Something happens →
- You respond automatically →
- Because you learned it earlier



Examples

- Phone notification sound → you check your phone

- Exam anxiety → you start biting nails
- Seeing a red traffic light → you stop the car

👉 These are **not inborn**. You **learned** them.

⚠ Important Point

Habits are formed not only by:

- **External stimuli** (sounds, sights, people)
but also by
- **Internal stimuli** (thoughts, feelings, anxiety, hunger)

Example:

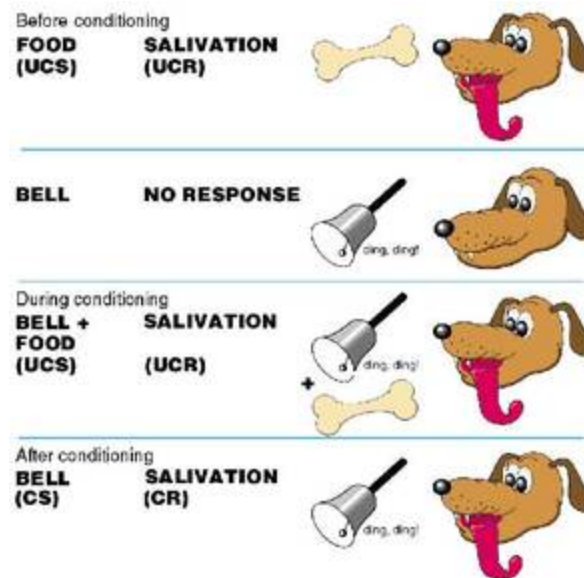
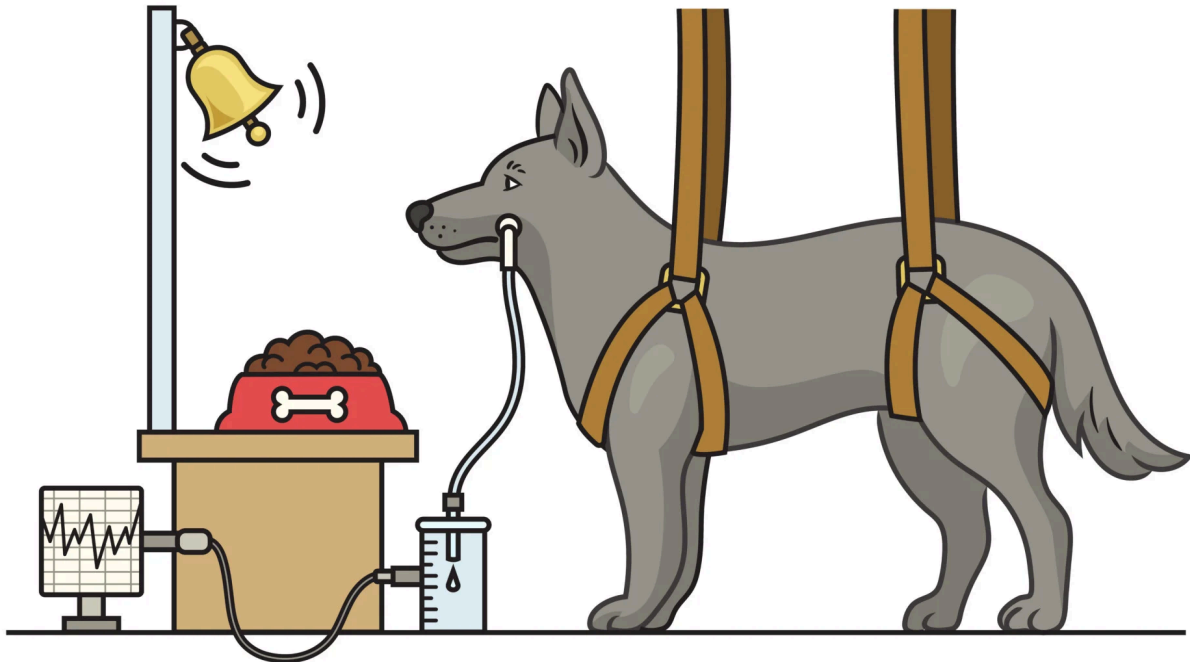
- Feeling anxious (internal stimulus) → avoiding people (response)
-

📖 BACKGROUND: WHERE DID THIS THEORY COME FROM?

Dollard and Miller didn't invent their ideas from nothing. They built on **earlier learning theorists**.

🧪 1. Pavlov – Classical Conditioning

Pavlov's experiment



Scientific Terms Explained

Term	Meaning
Unconditioned Stimulus (US)	Naturally causes a response (food)
Unconditioned Response (UR)	Natural response (salivation)
Conditioned Stimulus (CS)	Neutral thing that becomes meaningful (bell)
Conditioned Response (CR)	Learned response (salivation to bell)

Example

- Food → salivation (natural)
- Bell + food → salivation
- Bell alone → salivation (learned)

👉 Learning = **forming an S-R connection**

2. Watson – Behaviorism

Watson said:

| Psychology should study observable behavior, not thoughts or feelings.

- Focused on **overt behavior**
- Ignored mind and consciousness

Dollard & Miller accepted this focus on **behavior**, but they later included **internal stimuli** too.

3. Thorndike – Reward & Punishment

Thorndike discovered:

| Behaviors followed by rewards get stronger.

This is called the **Law of Effect**.

Example:

- Study hard → good grades → study habit strengthens
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4. Skinner – Operant Conditioning

Skinner focused on:

- **Behavior → consequences**
- **Reward strengthens behavior**

Successive Approximations (Very Important)

Definition:

Learning complex behavior by reinforcing **small steps** toward the final behavior.



Examples

- Learning to drive:
 - Sit in car → rewarded
 - Start engine → rewarded
 - Drive slowly → rewarded
 - Teaching a child to eat:
 - Hold spoon → rewarded
 - Bring to mouth → rewarded
-

WHAT IS LEARNING (According to Dollard & Miller)?

Definition

Learning is the process by which:

| A stimulus (cue) and a response become connected through reinforcement.

In plain English

Learning happens when:

- Something motivates you

- You notice a cue
 - You act
 - You get a reward
-

FOUR CONDITIONS FOR LEARNING (MEMORIZE THIS)

1 Drive – “I want something”

A **drive** is an internal state that pushes you to act.

Examples:

- Hunger
 - Thirst
 - Anxiety
 - Desire to succeed
-

2 Cue (Stimulus) – “I notice something”

A **cue** is a signal that tells you **how to respond**.

Examples:

- Exam timetable
 - Teacher’s expression
 - Traffic light
-

3 Response – “I do something”

This is your **behavior**.

Examples:

- Studying
- Eating

- Avoiding a situation
-

Reward (Reinforcement) – “I get something”

A **reward** strengthens the habit.

Examples:

- Food
 - Praise
 - Relief from anxiety
-

Example (Exam-friendly)

- **Drive:** Desire to pass exam
- **Cue:** Seeing books
- **Response:** Studying
- **Reward:** Good grades

 Habit forms

IMPORTANT RULE

For a habit to form:

- Stimulus & response must occur **close in time**
 - Response must be **rewarded**
-

CORE CONCEPTS OF THE THEORY

The Structure of Personality

Key Question

What makes a person **stable and consistent**?

✓ Answer

Habits

Explanation

Dollard & Miller didn't care much about:

- Traits
- Types
- Structures like id/ego/superego

They believed:

| Personality = collection of learned habits

The Dynamics of Personality (Motivation)

Key Term: Drive

A **drive** is an internal force that **pushes behavior**.

Primary Drives (Inborn)

- Hunger
 - Thirst
 - Pain
 - Sex
-

Secondary Drives (Learned) – VERY IMPORTANT

Definition:

Secondary drives are **learned motives** that develop from primary drives.

Examples:

- Anxiety

- Shame
- Guilt
- Desire to please others

📌 In modern life, **secondary drives control most behavior**, not hunger or thirst.

Example

- Child feels hunger (primary drive)
- Mother feeds child and smiles
- Smile becomes rewarding
- Later → mother's smile motivates behavior

➡ Smile becomes **secondary reward**

Secondary Rewards

Definition

Rewards that were **once neutral** but became rewarding through learning.

Examples:

- Praise
- Money
- Grades
- Approval

⚠ These rewards stay effective only if:

- Sometimes paired with **primary rewards**
-

3 The Development of Personality

Personality develops through **learning across life**.

i **Innate Equipment**

What you are born with:

- Reflexes
 - Basic drives
 - Nervous system
-

ii **The Learning Process**

Through:

- Reinforcement
 - Habit formation
 - S-R connections
-

iii **Secondary Drives & Learning**

Most adult behavior is motivated by:

- Anxiety
 - Social approval
 - Fear of failure
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iv **Higher Mental Processes**

Dollard & Miller explained thinking as:

| Internal S-R processes

Example:

- Thinking = silent talking
 - Planning = imagined responses
-

v **Social Context**

Learning happens in **society**:

- Family
- School
- Culture

Society teaches:

- What is rewarded
 - What is punished
-

vi Critical Stages of Development

Certain habits learned early:

- Are **hard to change**
 - Shape adult personality
-



Psychotherapy (Brief Idea)

Mental problems = **bad habits**

Therapy = **unlearning old habits + learning new ones**



Research

Their theory encouraged:

- Laboratory experiments
 - Scientific study of behavior
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SUMMARY (In One Breath)

- Personality = habits
- Habits = learned S-R connections

- Learning needs drive, cue, response, reward
 - Secondary drives (anxiety, approval) control adult behavior
 - Therapy works by changing habits
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EVALUATION (Quick Exam Points)

✓ Strengths

- Scientific
- Clear
- Useful in therapy

✗ Weaknesses

- Ignores emotions and creativity
 - Too mechanical
 - Underestimates free will
-

1. What is the Stimulus–Response (S–R) Theory of Dollard and Miller?

Answer:

The Stimulus–Response (S–R) Theory of Dollard and Miller is a **learning theory** which explains **personality as a collection of learned habits**. According to this theory, all behavior can be understood as a **link or association between a stimulus (cue) and a response (reaction)**. The theory emphasizes that **habits are formed, maintained, or extinguished** through learning, reinforcement, and experience, and that internal states such as drives and motives guide this learning process.

Explanation:

Dollard and Miller believed that the foundation of personality is **habit**, which is basically **learned behavior**. A habit forms when a person **notices a cue**

(stimulus), responds to it, and receives a reward, making it likely that the response will happen again. The theory combines ideas from **classical conditioning (Pavlov)**, **operant conditioning (Skinner)**, and **social learning**.

Example:

- A child touches a hot stove → feels pain → learns not to touch it again.
- Seeing a red light (stimulus) → stops the car (response) → avoids an accident (reward).

Tips:

- Remember **"S-R + Reward = Habit"**
 - Habits can be **external (observable)** or **internal (thoughts, feelings)**.
 - Focus on **how learning shapes personality**, not just traits.
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2. What is a habit according to Dollard and Miller?

Answer:

A habit is a **stable and learned connection between a stimulus (cue) and a response (behavior)**. It represents the **building block of personality**, and can involve both **external stimuli** (like sounds or sights) and **internal stimuli** (like thoughts, anxiety, or urges). Habits are formed through **reinforcement**, strengthened when a response produces a reward.

Explanation:

Habits are **not inborn**; they are learned. They become automatic over time because the connection between cue and response is **repeated and rewarded**. Dollard and Miller focus on **how habits are acquired, maintained, or replaced**, rather than listing all types of habits.

Example:

- External: Hearing a bell (stimulus) → salivating (response)
- Internal: Feeling nervous (stimulus) → biting nails (response)

Tips:

- Use the mnemonic: **H.A.B.I.T = Habit Always Builds If Trained**

- Remember: **Repetition + Reward = Habit Formation**
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3. What are primary and secondary drives?

Answer:

- **Primary drives** are **innate biological needs** that motivate behavior, such as hunger, thirst, pain avoidance, and sex.
- **Secondary drives** are **learned motives**, developed from primary drives, that guide complex behaviors. These include social or emotional motives like **anxiety, desire for approval, guilt, or shame**.

Explanation:

Primary drives exist from birth; they push a person to act to satisfy basic needs. Secondary drives emerge **through learning and social experience**, and often replace primary drives in influencing behavior. For example, a child learns that a **mother's smile** is rewarding, so the smile itself becomes a motivator for behavior, even without a primary reward.

Example:

- Primary: Hunger → eat food
- Secondary: Desire to please teacher → study for praise

Tips:

- Think: **Primary = biological, Secondary = learned or social**
 - Secondary drives often **hide the original primary drive** (e.g., studying instead of eating).
-

4. How does learning occur according to this theory?

Answer:

Learning occurs when a **stimulus (cue) and a response become connected through reinforcement**. For a behavior to be learned, four conditions must be met:

1. **Drive:** The person must want something.
2. **Cue (Stimulus):** The person must notice a signal for action.

3. **Response:** The person must act.
4. **Reward:** The action must be reinforced or rewarded.

Explanation:

Learning is essentially forming a **habit**, which is a **stable S–R connection**. The connection strengthens when the response is **rewarded**, and weakens or disappears if not reinforced. Both **external and internal stimuli** can trigger learning, and reinforcement can be **primary or secondary**.

Example:

- Drive: Desire to pass an exam
- Cue: Seeing books
- Response: Studying
- Reward: Good grades → habit of studying forms

Tips:

- Use the formula: **Drive + Cue + Response + Reward = Habit**
 - Remember: **Timing matters**—stimulus and response must occur close together for effective learning.
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5. What is successive approximations (shaping)?

Answer:

Successive approximations (also called shaping) is a **technique for teaching complex behaviors gradually**. It involves **reinforcing small steps toward the final behavior**, rather than expecting the behavior to occur all at once.

Explanation:

Instead of waiting for someone to perform the full behavior, you **reward every small step** in the right direction. Over time, these small reinforced steps build up into the full behavior.

Example:

- Teaching a child to feed themselves:

1. Hold spoon → reward
 2. Bring spoon to mouth → reward
 3. Eat independently → reward
- Learning to drive:
 1. Start engine → reward
 2. Drive in parking lot → reward
 3. Drive on road → reward

Tips:

- Remember: **Small steps → reinforcement → complex habit**
 - This technique is widely used in **child training, animal training, and skill learning**.
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6. How do secondary rewards work?

Answer:

Secondary rewards are **originally neutral stimuli** that acquire **reward value** through repeated association with **primary rewards**. They can **reinforce behavior by themselves**, but their effectiveness depends on occasional pairing with primary rewards.

Explanation:

Humans and animals learn that some neutral things predict a reward. Over time, these neutral things themselves become motivating. In society, most of our rewards are secondary (praise, money, grades), not primary (food, water).

Example:

- Mother's smile → paired with feeding → smile becomes rewarding
- Money → paired with buying food → money becomes motivating

Tips:

- Secondary rewards = **social and learned incentives**
- Important for **understanding human motivation** beyond basic drives.

7. What role does society play in habit formation?

Answer:

Society provides the **context in which habits are learned**, through rules, expectations, reinforcement, and social cues. Social approval, disapproval, and cultural norms act as **secondary reinforcers** that guide personality development.

Explanation:

Dollard and Miller emphasized that personality **develops in a social environment**. Children and adults learn which behaviors are rewarded or punished by **family, teachers, peers, and society**. This explains why culture influences behavior.

Example:

- Saying "thank you" is reinforced by praise → becomes habitual polite behavior
- Avoiding lying → reinforced by punishment → develops honesty

Tips:

- Social environment = **source of secondary drives**
- Think of society as the **"training ground" for personality habits**

8. How can this theory be applied to psychotherapy?

Answer:

Psychotherapy, according to Dollard and Miller, focuses on **changing maladaptive habits**. Mental health problems are seen as **bad or maladaptive habits**, and therapy aims to **extinguish old habits and reinforce new, adaptive habits**.

Explanation:

- Problem behaviors often stem from **maladaptive S-R connections** (e.g., fear, anxiety, avoidance).
- Therapy uses **learning principles**, such as reinforcement, modeling, and shaping, to **replace these with healthier behaviors**.

Example:

- Anxiety → avoidance (maladaptive habit)
- Therapy → exposure to feared stimulus → reinforced calm response → habit replaced

Tips:

- Therapy = **“habit remodeling”**
 - Focus on **small, reinforced steps** (successive approximations)
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A. True/False Statements

1. **Dollard and Miller believed personality is primarily the result of inherited traits rather than learned habits.**

Answer: False

Explanation: Dollard & Miller emphasized that **personality is a collection of learned habits**. Traits are not the main focus; habits, formed through learning and reinforcement, determine personality.

2. **A habit is defined as a stable connection between a stimulus and a response.**

Answer: True

Explanation: By definition, a **habit = S–R connection**. It can involve both **external** (observable) and **internal** (thoughts, feelings) stimuli.

3. **Secondary drives are innate and present at birth.**

Answer: False

Explanation: Secondary drives are **learned motives** (e.g., anxiety, desire to please) that develop from **primary drives** and social reinforcement.

4. **Successive approximations (shaping) is used to teach complex behaviors gradually.**

Answer: True

Explanation: This technique **reinforces small steps toward the final behavior**, widely used in skill learning, therapy, and animal training.

5. **According to Dollard and Miller, internal thoughts and feelings cannot form S-R habits.**

Answer: False

Explanation: S-R habits **can form with internal stimuli** such as anxiety, fear, or desire, not only with external cues.

B. Conceptual MCQs

1. Which of the following best represents the four conditions for learning in Dollard & Miller's theory?

- A. Stimulus, Trait, Personality, Reward
- B. Drive, Cue, Response, Reward
- C. Habit, Reward, Punishment, Drive
- D. Motivation, Drive, Response, Personality

Answer: B. Drive, Cue, Response, Reward

Explanation: Learning occurs when a **drive motivates** the subject, a **cue signals** the response, a **response occurs**, and it is followed by a **reward**.

1. Secondary rewards:

- A. Are always primary biological reinforcers
- B. Are neutral stimuli that acquire reinforcing value through learning
- C. Cannot reinforce behavior by themselves
- D. Are innate and inborn

Answer: B. Are neutral stimuli that acquire reinforcing value through learning

Explanation: Secondary rewards (e.g., praise, money, approval) **become motivating because they were repeatedly paired with primary rewards**.

1. Which of the following statements is correct about primary and secondary drives?

- A. Primary drives are learned; secondary drives are innate

- B. Primary drives include anxiety and shame
- C. Secondary drives replace primary drives in modern society
- D. Secondary drives are irrelevant in behavior formation

Answer: C. Secondary drives replace primary drives in modern society

Explanation: In modern life, **learned social motives** often guide behavior more than basic biological drives.

1. What is the main goal of psychotherapy according to Dollard and Miller?

- A. Exploring unconscious conflicts
- B. Changing maladaptive S-R habits
- C. Reinforcing primary drives
- D. Eliminating personality traits

Answer: B. Changing maladaptive S-R habits

Explanation: Therapy focuses on **unlearning bad habits and reinforcing adaptive habits** using learning principles.

1. Which of the following is an example of internal stimulus forming a habit?

- A. Seeing a red traffic light → stopping
- B. Hearing a bell → salivating
- C. Feeling anxious → biting nails
- D. Receiving a candy → smiling

Answer: C. Feeling anxious → biting nails

Explanation: Internal states like anxiety can **trigger learned responses**, forming habits without external cues.

C. Situational / Application MCQs

1. Ahmed wants to motivate his student to study. He provides small rewards after each completed chapter. Which learning principle is Ahmed applying?

- A. Classical conditioning

- B. Successive approximations (shaping)
- C. Operant extinction
- D. Primary drive reinforcement

Answer: B. Successive approximations (shaping)

Explanation: Ahmed is **reinforcing small steps toward the final goal**, which is **exactly shaping**.

1. Sara feels anxious every time she speaks in public. Her teacher praises her when she attempts. Over time, she enjoys public speaking. This is an example of:

- A. Secondary drive formation
- B. Classical conditioning
- C. Habit extinction
- D. Punishment

Answer: A. Secondary drive formation

Explanation: Anxiety (initially a primary response) is **replaced by learned motivation** due to **praise (secondary reward)**.

1. A child touches a hot stove and feels pain. Later, they avoid touching the stove. In S-R theory terms:

- A. Hot stove = response; Pain = stimulus
- B. Hot stove = stimulus; Avoiding = response; Pain = reward
- C. Pain = drive; Stove = response
- D. Stove = reward; Avoidance = drive

Answer: B. Hot stove = stimulus; Avoiding = response; Pain = reward

Explanation: The stove acts as **cue (stimulus)**, avoidance is the **response**, and the pain serves as **primary reinforcement** teaching the habit.

1. Which of the following demonstrates secondary reinforcement in society?

- A. Food for hunger

- B. Money earned from work
- C. Pain from injury
- D. Water when thirsty

Answer: B. Money earned from work

Explanation: Money **is not biologically needed** but acquires value through **association with primary rewards** like food, comfort, and security.

1. A person learns to avoid lying because their parents scold them every time they lie. Which principle is illustrated?

- A. Secondary drive formation
- B. Punishment as reinforcement
- C. Successive approximations
- D. Classical conditioning

Answer: B. Punishment as reinforcement

Explanation: Behavior is reduced through **negative consequences** (scolding), which is **a key part of S-R learning**.

Cheat Sheet

1 Key Terms & Definitions

Term	Definition	Example	Memory Hook / Mnemonic
Habit	Stable learned connection between stimulus (cue) and response (behavior)	Hearing a bell → salivating	H.A.B.I.T = Habit Always Builds If Trained
Stimulus (Cue)	Signal or event that triggers a response	Red traffic light	S = Signal
Response	Behavior triggered by a stimulus	Stop car at red light	R = Reaction

Term	Definition	Example	Memory Hook / Mnemonic
Drive	Internal state motivating behavior	Hunger, desire to succeed	D = Desire to act
Reward / Reinforcement	Positive outcome that strengthens habit	Good grades, mother's smile	R = Reward → habit grows
Primary Drive	Innate, biological needs	Hunger, thirst, pain avoidance	P = Primary = Physiological
Secondary Drive	Learned motives derived from primary drives	Desire for approval, anxiety, shame	S = Social / Secondary
Secondary Reward	Originally neutral event that becomes motivating via learning	Money, praise, grades	S = Secondary → Learned reward
Successive Approximations / Shaping	Gradual reinforcement of steps toward final behavior	Teaching child to feed themselves	Step-by-step shaping

2 Four Conditions for Learning

Formula: Drive + Cue + Response + Reward = Habit

Step	Meaning	Example	Tip
Drive	Motivation to act	Desire to pass exam	Ask: "Do they want it?"
Cue (Stimulus)	Signal to respond	Seeing books on desk	Ask: "What triggers action?"
Response	The action itself	Studying	"What do they do?"
Reward	Outcome that reinforces habit	Good grades	"Was the action reinforced?"

3 Structure of Personality

- Personality = **collection of habits**

- Focus on **how habits form**, not traits
- Habits can be **external (observable)** or **internal (thoughts, feelings)**

Tip: Think “**Personality = Library of Habits**”

4 Dynamics of Personality (Motivation)

- **Primary drives** → innate biological needs
- **Secondary drives** → learned from experience & society
- Secondary drives often **replace primary drives** in modern behavior
- Most adult behavior is guided by **secondary drives** (anxiety, desire to please, shame)

Example:

- Baby's hunger → mother feeds + smiles → mother's smile becomes motivating (secondary drive/reward)

Mnemonic: “**Primary Needs → Secondary Wants**”

5 Learning & Habit Formation

Key Points:

- Learning = **forming S-R connections through reinforcement**
- Timing matters: **Stimulus + Response must be close in time**
- Reward strengthens connection
- Internal & external cues can trigger learning

Example:

- Drive: Want to pass
- Cue: Seeing books
- Response: Study
- Reward: Good grades → Habit formed

Mnemonic: "D.C.R.R = Drive → Cue → Response → Reward"

6 Successive Approximations (Shaping)

- Reinforce **small steps** toward final behavior
- Gradual learning of **complex behavior**

Examples:

- Child learning to feed → Hold spoon → Bring to mouth → Eat independently
- Learning to drive → Sit in car → Start engine → Drive slowly → Drive independently

Tip: Think "**Little wins build big habits**"

7 Secondary Rewards

- Neutral stimuli acquire **reward value through repeated pairing with primary reward**
- Reinforce behavior **even without primary reward** occasionally
- Example: Money, praise, grades

Memory Hook: "Neutral → Reward via Learning"

8 Social Context

- Habits develop within **family, culture, society**
- Social reinforcement = **primary source of secondary drives**
- Behaviors rewarded socially are **more likely to be repeated**

Example:

- Saying "please" and "thank you" → praised → becomes habitual
- Avoiding lying → punished → becomes honest habit

Mnemonic: "Society = Habit Trainer"

9 Psychotherapy Application

- Mental problems = **maladaptive S-R habits**
- Therapy = **unlearning bad habits + learning new adaptive habits**
- Uses **reinforcement, shaping, and modeling**

Example:

- Anxiety → avoidance → therapy reinforces calm response → habit replaced

Memory Hook: "Therapy = Habit Remodeling"

10 Table

Concept	Key Point	Example	Mnemonic
Habit	S-R connection	Bell → salivation	H.A.B.I.T
Drive	Motivates action	Hunger, success	D = Desire
Cue / Stimulus	Triggers response	Red light	S = Signal
Response	Behavior	Stop car	R = Reaction
Reward / Reinforcement	Strengthens habit	Grades, praise	R → Reinforce
Primary vs Secondary	Innate vs learned	Hunger vs Approval	"Primary Needs → Secondary Wants"
Secondary Reward	Learned incentive	Money, smile	Neutral → Learned reward
Shaping	Gradual steps	Teaching child	Step-by-step shaping

Summary

- **H.A.B.I.T** = Habit Always Builds If Trained
- **D.C.R.R** = Drive → Cue → Response → Reward
- **Primary Needs → Secondary Wants**
- **Society = Habit Trainer**
- **Therapy = Habit Remodeling**